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https://www.tensorflow.org/api_docs/python/tf/compat/v1/cond

> *Return true_fn() if the predicate pred is true else false_fn(). (deprecated arguments)*

Function Signatures

```
true_fn()
false_fn()
tf.compat.v1.cond( pred, true_fn=None, false_fn=None, strict=False, name=None, fn1=None, fn2=None )
(fn1, fn2)
```

Code Examples

```
tf.compat.v1.cond(
pred,
true_fn=None,
false_fn=None,
strict=False,
name=None,
fn1=None,
fn2=None
)
```

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pred,
true_fn=None,
false_fn=None,
strict=False,
name=None,
fn1=None,
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z = tf.multiply(a, b)
result = tf.cond(x < y, lambda: tf.add(x, z), lambda: tf.square(y))
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x = tf.constant(2)
y = tf.constant(5)
def f1(): return tf.multiply(x, 17)
def f2(): return tf.add(y, 23)
r = tf.cond(tf.less(x, y), f1, f2)
# r is set to f1().
# Operations in f2 (e.g., tf.add) are not executed.
```


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y = tf.constant(5)
def f1(): return tf.multiply(x, 17)
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```


Documentation

Return true_fn() if the predicate pred is true else false_fn(). (deprecated arguments)

true_fn and false_fn both return lists of output tensors. true_fn and false_fn must have the same non-zero number and type of outputs.

Although this behavior is consistent with the dataflow model of TensorFlow, it has frequently surprised users who expected a lazier semantics. Consider the following simple program:

If x < y, the tf.add operation will be executed and tf.square operation will not be executed. Since z is needed for at least one branch of the cond, the tf.multiply operation is always executed, unconditionally.

Note that cond calls true_fn and false_fn exactly once (inside the call to cond, and not at all during Session.run()). cond stitches together the graph fragments created during the true_fn and false_fn calls with some additional graph nodes to ensure that the right branch gets executed depending on the value of pred.

tf.cond supports nested structures as implemented in tensorflow.python.util.nest. Both true_fn and false_fn must return the same (possibly nested) value structure of lists, tuples, and/or named tuples. Singleton lists and tuples form the only exceptions to this: when returned by true_fn and/or false_fn, they are implicitly unpacked to single values. This behavior is disabled by passing strict=True.

Returns

Raises

Example: